

Alternating Current and Electromagnetic Waves, solved

Step-by-step solutions to every practice problem in Chapter 21. Try each problem on your own before you read its solution.

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P21.1

The *useful* effect of a current, such as heating or doing work, depends on the average of $i^2 R$, not on the peak. The rms value is the steady DC value that would deliver the same average power. Quoting rms values lets the familiar DC formulas ($P = V_{\text{rms}} I_{\text{rms}}$ for a resistor) be used unchanged. A 120 V rms supply has peaks of $120\sqrt{2} = 170$ V.

P21.2

The capacitive reactance is $X_C = 1/(\omega C)$.

- For DC, $\omega = 0$, so $X_C \rightarrow \infty$. Once the capacitor has charged up, no further current flows: it blocks DC.
- For AC, the capacitor continually charges and discharges, so current flows back and forth in the circuit, although no charge ever crosses the gap.
- At **higher frequencies**, the capacitor reverses before it can build up much charge, so its voltage stays small for a given current. That means lower reactance and a larger current.

P21.3

At resonance $X_L = X_C$, so $Z = R$ (the minimum possible impedance) and $\phi = 0$: the current is in phase with the source voltage.

$V_L = IX_L$ and $V_C = IX_C$ are equal in size but 180° out of phase, so they cancel in the sum. Each one separately can be large. At resonance, $I = V/R$, so

$$V_L = V_C = \frac{V}{R}X_L = QV,$$

which is Q times the source voltage. With $Q = 50$, a 10 V source can produce 500 V across the inductor. The energy sloshing between L and C builds up over many cycles, exactly as a swing pushed at its natural frequency builds up a large amplitude.

P21.4

In empty space, $\oint \vec{E} \cdot d\vec{A} = 0$ and $\oint \vec{B} \cdot d\vec{A} = 0$. For a plane wave travelling along x , apply these to a box with faces perpendicular to x . Any field component along x would have to be the same on the front and back faces at every moment, so it could not vary as a wave. A longitudinal wave is therefore impossible. Faraday's law and the Ampère–Maxwell law then require $\vec{E} \perp \vec{B}$.

Direction of travel: along $\vec{E} \times \vec{B} = \hat{j} \times \hat{k} = \hat{i}$, that is, the $+x$ direction.

P21.5

$$X_L = 2\pi fL = 2\pi(50)(0.20) = 63 \, \Omega, \quad I_{\text{rms}} = \frac{230}{62.8} = 3.7 \, \text{A}.$$

Average power: zero. The voltage leads the current by 90° , so $\cos \phi = 0$. Energy is stored in the field during one quarter-cycle and returned during the next.

P21.6

$$X_C = \frac{1}{2\pi(50)(50 \times 10^{-6})} = 64 \, \Omega, \quad I_{\text{rms}} = \frac{230}{63.7} = 3.6 \, \text{A}.$$

Doubling the frequency halves X_C , so the current doubles, to 7.2 A.

P21.7**Reactances:**

$$X_L = 2\pi(60)(0.10) = 37.7 \, \Omega, \quad X_C = \frac{1}{2\pi(60)(50 \times 10^{-6})} = 53.1 \, \Omega.$$

Impedance and current:

$$Z = \sqrt{20^2 + (37.7 - 53.1)^2} = \sqrt{400 + 236} = 25.2 \, \Omega, \quad I = \frac{120}{25.2} = 4.76 \, \text{A}.$$

Phase: $\tan \phi = -15.4/20$, so $\phi = -37.5^\circ$. The circuit is capacitive, and the current leads.**Power:** the power factor is $\cos \phi = 0.79$, and

$$\bar{P} = I^2 R = (4.76)^2(20) = 453 \, \text{W}.$$

Resonance:

$$f_0 = \frac{1}{2\pi\sqrt{(0.10)(50 \times 10^{-6})}} = 71 \, \text{Hz}.$$

P21.8**(a) Real and reactive power.**

$$P = VI \cos \phi = 230(10)(0.70) = 1610 \, \text{W}.$$

$$\sin \phi = \sqrt{1 - 0.70^2} = 0.714, \quad Q_{\text{reactive}} = VI \sin \phi = 1640 \, \text{VAR (inductive)}.$$

(b) Capacitor. It must supply an equal and opposite reactive power, $V^2/X_C = V^2\omega C$:

$$C = \frac{Q_{\text{reactive}}}{V^2\omega} = \frac{1640}{(230)^2(314)} = 9.9 \times 10^{-5} \, \text{F} \approx 99 \, \mu\text{F}.$$

(c) New current. With a power factor of 1, the supply provides only the real power:

$$I = \frac{1610}{230} = 7.0 \, \text{A},$$

down from 10 A. The motor runs exactly as before, but the cables and the transformer carry 30% less current, so their $I^2 R$ losses are about halved.

P21.9

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(0.025)(4.0 \times 10^{-6})}} = 3.16 \times 10^3 \text{ rad/s}, \quad f = 503 \text{ Hz.}$$

$$Q_0 = CV = (4.0 \times 10^{-6})(50) = 2.0 \times 10^{-4} \text{ C}, \quad I_0 = \omega_0 Q_0 = 0.63 \text{ A.}$$

Check by energy: $\frac{1}{2}CV^2 = 5.0 \text{ mJ}$ and $\frac{1}{2}LI_0^2 = \frac{1}{2}(0.025)(0.632)^2 = 5.0 \text{ mJ. } \checkmark$

P21.10

$$C = 1/(\omega^2 L).$$

- At 88 MHz: $\omega = 5.53 \times 10^8 \text{ rad/s}$, so $C = 1/[(5.53 \times 10^8)^2(0.30 \times 10^{-6})] = 10.9 \text{ pF}$.
- At 108 MHz: $\omega = 6.79 \times 10^8 \text{ rad/s}$, so $C = 7.2 \text{ pF}$.

The capacitance must range from about 7.2 pF to 10.9 pF. A modest change of about 50% covers the whole band, because $f \propto C^{-1/2}$.

P21.11

Intensity:

$$I = \frac{P}{\pi r^2} = \frac{5.0 \times 10^{-3}}{\pi(0.50 \times 10^{-3})^2} = 6.4 \times 10^3 \text{ W/m}^2,$$

about six times bright sunlight.

Peak field:

$$E_0 = \sqrt{\frac{2I}{c\epsilon_0}} = \sqrt{\frac{2(6.4 \times 10^3)}{2.66 \times 10^{-3}}} = 2.2 \times 10^3 \text{ V/m.}$$

Retinal danger. The beam is coherent and nearly parallel, so the eye's lens focuses it to a spot only about 10–20 μm across on the retina, which concentrates it 1000–10 000-fold. The local intensity can then reach 10^7 – 10^8 W/m^2 , enough to burn retinal tissue within a fraction of a second. A diffuse source of the same power, such as a lamp, cannot be focused to such a small spot.

P21.12

(a) **Wavelength:**

$$\lambda = \frac{c}{f} = \frac{3.0 \times 10^8}{2.45 \times 10^9} = 0.122 \text{ m} = 12.2 \text{ cm}.$$

(b) **Hot spots.** The antinodes of a standing wave are $\lambda/2 \approx 6.1$ cm apart. To measure c , remove the turntable, heat a bar of chocolate briefly, and measure the distance d between neighbouring melted spots. Then $\lambda = 2d$ and $c = 2df$, using the frequency printed on the oven. The result is typically within a few percent of 3×10^8 m/s.

(c) **The mesh.** A conducting mesh whose holes are much smaller than the wavelength (2 mm against 122 mm) behaves almost like a solid metal sheet, and reflects the microwaves. Visible light has a wavelength of about $0.5 \mu\text{m}$, thousands of times smaller than the holes, so it passes straight through. That is why you can see the food but are protected from the microwaves.

P21.13

(a) **Current.** The impedance is $Z = \sqrt{R^2 + (X_L - X_C)^2}$, with $X_L = \omega L$ and $X_C = 1/\omega C$, and $I = V/Z$. ✓

(b) **Half power.** $\bar{P} = I^2 R$ is proportional to $1/Z^2$. It falls to half its resonant value when $Z^2 = 2R^2$, that is, when $|\omega L - 1/\omega C| = R$. Near resonance, write $\omega = \omega_0 + \delta$:

$$\omega L - \frac{1}{\omega C} = \frac{L}{\omega}(\omega^2 - \omega_0^2) \approx 2L\delta.$$

The half-power points are at $\delta = \pm R/2L$, so the full width is

$$\Delta\omega = \frac{R}{L} = \frac{\omega_0}{Q}.$$

(c) **Radio tuner.**

$$Q = \frac{\omega_0 L}{R} = \frac{2\pi(10^8)(2.0 \times 10^{-6})}{5.0} = 251, \quad \Delta f = \frac{f_0}{Q} = \frac{10^8}{251} = 400 \text{ kHz}.$$

The bandwidth spans about two channel spacings (200 kHz), so this circuit alone cannot reliably separate neighbouring stations. Real receivers convert the signal to a fixed *intermediate frequency* (10.7 MHz for FM), where sharper filters can be used.

P21.14

(a) **Displacement current.** Between the plates, $E = \sigma/\epsilon_0 = Q/(\epsilon_0\pi R^2)$, so the total electric flux is $\Phi_E = EA = Q/\epsilon_0$. Then

$$I_d = \epsilon_0 \frac{d\Phi_E}{dt} = \frac{dQ}{dt} = I.$$

The displacement current in the gap equals the conduction current in the wires, so “current” is continuous through the capacitor.

(b) **Field between the plates.** Take a circular loop of radius $r < R$, coaxial with the plates. It encloses a fraction r^2/R^2 of the electric flux:

$$B(2\pi r) = \mu_0\epsilon_0 \frac{d\Phi_E}{dt} \cdot \frac{r^2}{R^2} = \mu_0 I \frac{r^2}{R^2} \Rightarrow B = \frac{\mu_0 I r}{2\pi R^2}.$$

This is the same as the field inside a wire carrying uniform current, and it has been measured experimentally.

P21.15

(a) **Faraday’s law on a small rectangle.** Take a rectangle in the xy -plane with width dx (along x) and height h (along y). $\vec{E}$ points along y , so only the two vertical sides contribute to the line integral (going anticlockwise):

$$\oint \vec{E} \cdot d\vec{l} = [E(x+dx) - E(x)]h = \frac{\partial E}{\partial x} h dx.$$

The flux of $\vec{B}$ (along z) through the rectangle is $Bh dx$, so Faraday’s law gives

$$\frac{\partial E}{\partial x} = -\frac{\partial B}{\partial t}.$$

Substituting the wave:

$$kE_0 \cos(kx - \omega t) = \omega B_0 \cos(kx - \omega t) \Rightarrow E_0 = \frac{\omega}{k} B_0 = cB_0. \checkmark$$

(b) **Equal energy densities.**

$$u_B = \frac{B^2}{2\mu_0} = \frac{E^2}{2\mu_0 c^2} = \frac{\epsilon_0 E^2}{2} = u_E,$$

using $c^2 = 1/(\mu_0\epsilon_0)$. The two fields carry exactly equal shares of the wave’s energy at every point and every instant.

P21.16

(a) **Pressure and acceleration** (perfect reflection doubles the momentum transfer):

$$p = \frac{2I}{c} = \frac{2(1360)}{3.0 \times 10^8} = 9.1 \times 10^{-6} \text{ Pa.}$$

$$a = \frac{p}{\sigma} = \frac{9.07 \times 10^{-6}}{0.010} = 9.1 \times 10^{-4} \text{ m/s}^2.$$

(b) **Comparison:** this is only about 15% of the Sun's gravitational pull ($5.93 \times 10^{-3} \text{ m/s}^2$). Gravity still dominates, but the sail can still steer: tilting it changes the sail's orbital energy, so it can spiral slowly inward or outward.

(c) **Hovering:** the radiation force per unit area must balance gravity:

$$\sigma_{\max} = \frac{2I}{cg_{\odot}} = \frac{9.07 \times 10^{-6}}{5.93 \times 10^{-3}} = 1.5 \times 10^{-3} \text{ kg/m}^2 = 1.5 \text{ g/m}^2.$$

Independent of distance: both sunlight's intensity and the Sun's gravity fall off as $1/r^2$, so their ratio is the same at every distance from the Sun. A sail light enough to hover at 1 AU would hover anywhere. Current sail films (a few μm of aluminised polymer) weigh several g/m^2 , which is close to this limit.

P21.17

(a) **Absorbed power:** $P = 2.0 \text{ W/kg} \times 70 \text{ kg} = 140 \text{ W}.$

(b) **Temperature rise with no cooling:**

$$\Delta T = \frac{Pt}{mc} = \frac{140(600)}{70(3500)} = 0.34 \text{ K}.$$

In practice the body's thermoregulation (blood flow and sweating) removes much of this heat, so the actual rise is smaller.

(c) **Local heating.** The SAR limit is a *whole-body average*. RF fields are strongly concentrated near the tips and edges of metal implants and wires, which act like small antennas. Induced currents also flow round any closed conducting loop, such as hands clasped together, thighs touching, or ECG leads lying in a loop. Where these currents are squeezed through a small contact area, the current density, and so the local heating (J^2/σ), can be very large. Patients have suffered skin burns this way. Hence the protocols: no crossed limbs, insulating pads, and leads routed straight and away from the body.

P21.18

(a) **Required inductance:**

$$L = \frac{1}{\omega^2 C} = \frac{1}{(2\pi \times 50)^2 (10 \times 10^{-6})} = \frac{1}{(9.87 \times 10^4)(10^{-5})} = 1.0 \text{ H.}$$

(b) **Practicality.** A 1 H inductor is large and heavy (an iron-cored coil), has significant resistance, which lowers Q and broadens the notch, drifts with temperature, and can itself pick up mains hum magnetically. It would also distort the ECG near 50 Hz. Modern instruments therefore use **active filters**, built from op-amps, resistors and capacitors, which give sharp, stable responses without inductors. Most now use **digital filtering** after digitisation. Prevention is better than cure:

- good, low-impedance electrode contact keeps the two inputs balanced;
- a high common-mode rejection ratio in the amplifier ignores interference that appears equally on both inputs;
- a **driven right-leg** circuit actively feeds back an inverted copy of the common-mode voltage into the patient, cancelling much of the hum at its source.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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